

Master's Programme in Computer, Communication and Information Sciences

Proof of concept of centralized personal finance application

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Abstract

This thesis presents the design and development of a centralized personal finance management application using the Design Science Research Methodology (DSRM). The work is based in prior research on financial behavior, planning, and management tools, explaining how motivation, structured planning, and digital solutions influence financial well-being. The goal was to address real-world challenges faced by an individual stakeholder, whose existing financial tracking system relied on fragmented tools and manual processes. A proof-of-concept (PoC) application was developed using Flutter, .NET, and MongoDB, including functionalities such as asset tracking, categorized cash flow visualization, real-time data updates, and financial report generation. The solution was iteratively designed based on stakeholder's feedback and evaluated through task-based testing, structured questionnaires, and comparison to the baseline system. Results show significant improvements in automation, accuracy, and usability. While some features, such as authentication and full banking integration, were out of scope for this PoC, the system provides a strong foundation for further development and wider applicability in the domain of personal finance management.

Keywords Personal finance, Financial planning, Design Science Research, Proof of Concept, Personal finance application, Financial tracking, User-centered design

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Tiivistelmä

Tämä diplomityö käsittelee keskitetyn henkilökohtaisen taloudenhallintasovelluksen suunnittelua ja kehittämistä hyödyntäen Design Science Research Methodology (DSRM) -tutkimusmenetelmää. Työn teoreettinen tausta pohjautuu aiempaan kirjallisuuteen taloudellisesta käyttäytymisestä, suunnittelusta ja taloudenhallintatyökaluista, joissa korostuvat motivaatio, suunnitelmallisuus ja digitaaliset ratkaisut taloudellisen hyvinvoinnin tukena. Työn tavoitteena oli parantaa yksityisen toimijan olemassa olevaa järjestelmää, joka koostui hajanaisista työkaluista ja manuaalisista prosesseista. Konseptitodistus kehitettiin hyödyntäen teknologioita kuten Flutter, .NET ja MongoDB. Sovellus sisältää toimintoja kuten varallisuuden seuranta, kategorisoitu kassavirta-analyysi, reaaliaikainen datan päivitys sekä talousraporttien generointi. Ratkaisu suunniteltiin iteratiivisesti toimijan palautteen perusteella ja arvioitiin tehtäväpohjaisella testauksella, strukturoidulla kyselyllä ja vertailulla aiempaan järjestelmään. Tulokset osoittavat merkittäviä parannuksia automaatiossa, luotettavuudessa ja käytettävyydessä. Vaikka kaikkia toimintoja, kuten tunnistautumista tai pankki-integraatioita, ei toteutettu konseptitodistuksessa, sovellus luo vahvan pohjan jatkokehitykselle ja laajemmalle käytettävyydelle henkilökohtaisen taloudenhallinnan alueella.

Avainsanat Henkilökohtainen talous, Taloussuunnittelu, Kehittämistutkimus, Konseptitodistus, Henkilökohtaisen talouden sovellus, Talouden seuranta, Käyttäjälähtöinen suunnittelu

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Symbols and abbreviations

Abbreviations

API	Application Programming Interface
DB	Database
DS	Design Science
DSR	Design Science Research
DSRM	Design Science Research Methodology
FR	Functional Requirement
PFM	Personal Finance Management
PoC	Proof-of-concept
QA	Quality Attribute
STD	Self-Determination Theory
UI	User Interface
UX	User Experience

1 Introduction

1.1 Context and motivation

Peter Drucker said, "If you can't measure it, you can't manage it" highlighting the importance of data in decision-making. Hamming's quote, "You get what you measure" represents how we guide ourselves with measuring the steps we take toward achieving our goals. W. Edwards Deming presented the saying "In God we trust, all others bring data" which is a reminder that we should base our decisions off data and not assumptive beliefs.

This holds true for Personal Finance Management (PFM) which requires financial literacy, motivation and structured planning to make informed financial decisions. Studies suggest that people with higher rates of financial literacy are more likely to participate in types of behaviors related to saving, investing, and avoiding unnecessary financial mistakes (Lusardi and Mitchell 2014; Rajesh and Giridhar 2024). In addition, global surveys suggest that lower rates of financial literacy remain in many demographics, particularly young adults, women, and among lower income or education levels (Atkinson and Messy 2012; Lusardi and Mitchell 2014).

Financial knowledge alone is not enough to create behavioral change. Di Domenico et al. (2022) explains that autonomous motivation, driven by internal values and interest, is more strongly associated with sustainable financial habits and well-being than controlled motivation which relates to external pressure or guilt. Furthermore, Kaye et al. (2014) has shown that financial management behaviors are driven by emotional, social and experiential motivations, not just rational knowledge. Shafiee et al. (2024) suggests that well-designed, personalized financial tools can increase financial literacy and support decision-making, while still fulfilling an educational and advisory role.

While digital PFM tools aim to help users track expenses, manage budgets, and visualize financial data, their adoption remains limited. Stefanov et al. (2024) found that only few available tools were able to meet all consumer needs for easier financial management, especially in underrepresented regions. In addition, many tools only offer one service to the users, such as budgeting, and do not provide educational content which would help raise trust to financial services (Torno et al. 2021). Research has shown that adding personalization, gamified feedback, and simplified interactions can improve user engagement and financial self-efficacy (Bitrián et al. 2021; Cederberg 2013), yet these elements are still uncommon in mainstream applications.

This thesis responds to these limitations by designing a centralized personal finance application that consolidates financial data, supports user learning and motivation, and offers tools for financial planning and analysis.

1.2 Objectives and scope

The goal of this study is to design and build a proof-of-concept of centralized personal finance application that helps individuals make informed, data-driven financial decisions. Following the Design Science Research Methodology (DSRM), the study aims to identify user needs, create a customized solution, and evaluate its practical relevance and functionality. The application was developed for a private stakeholder with real-life requirements.

In particular, the application aims to achieve:

- Consolidation of financial data across asset classes including cash, shares and real estate from multiple sources on a single platform
- Track a user's net worth development over time through visualized data
- Provide clear visual overviews of transactions and cash flow through tables and charts
- Improve transparency and accessibility by structuring personal financial data in a scalable, easy-to-navigate system
- Provide breakdowns of the assets by categories and subcategories including market value, equity, country attribution and more

This work contributes to the development of practical PFM tools by focusing on data integration, visual clarity, and actionable insights. It also demonstrates the application of DSRM to guide artifact development and iteration based on stakeholder's feedback and evaluation.

1.3 Use of artificial intelligence tools

Artificial intelligence tools, including ChatGPT, were used to support the structuring and writing of this thesis. AI assistance was first used to help outline the overall structure of the thesis and deciding what content should be included in each chapter and section. Once the structure was in place, AI was used to help draft some paragraphs based on predefined ideas, bullet points, and notes.

In the Literature review (Chapter 2), the process involved first highlighting and collecting relevant information from selected academic sources. The material was manually organized into clearly defined subsections. AI tools were then used to help compose coherent and logically structured paragraphs, ensuring that the collected sources were presented under the appropriate themes and headings.

Additionally, AI was used throughout the thesis to check that statements were supported by the cited literature and to refine clarity, grammar, and academic tone. All AI-assisted content was reviewed, edited, and finalized to ensure accuracy, integrity,

and appropriateness for academic work.

1.4 Thesis structure

This thesis consists of ten chapters. Chapter 1 introduces the research context, motivation, and objectives. Chapter 2 reviews relevant literature on financial literacy, personal finance planning, management tools, and DSRM. Chapter 3 presents the research methodology. Chapters 4 to 8 follow the DSRM process: identifying the problem, defining solution objectives, designing and developing the proof-of-concept, demonstrating its use, and evaluating its effectiveness. Chapter 9 discusses the broader impact of the results and future development opportunities. Chapter 10 concludes the thesis by summarizing key findings and reflecting on the research outcomes.

2 Literature review

This chapter provides the theoretical background for the research by investigating the three topics relevant to the development of a centralized personal finance application. First, it looks into financial literacy and behavior, focusing how knowledge, motivation, and habits shape financial outcomes. Second, it looks into personal financial planning as a structured process aimed at achieving financial goals through budgeting, saving, and investing. Third, it looks into current state of personal finance management tools, highlighting current capabilities, user needs, and design opportunities. Finally, the chapter explains Design Science Research Methodology (DSRM), tracing its origins, artifact types, and structured research process, which forms the methodological base of this thesis.

2.1 Financial literacy and behaviors

Financial literacy is essential for managing money, making informed decisions, and achieving long-term financial well-being. Rajesh and Giridhar (2024) argue that low levels of financial literacy can lead to significant consequences related to the accumulation of overall wealth, such as unsustainable levels of debt, poor credit, and possibly even bankruptcy. Many people struggle with financial management, such as, overspending, making poor investment decisions, borrowing too much money unnecessarily, and failing to plan for retirement. They note that increased financial literacy allows individuals to have better saving and spending routines, control their debt better, and achieve improved financial strength. Likewise, Lusardi and Mitchell (2014) confirm that low levels of financial literacy are present in all populations, particularly the elderly and youth. They say that all populations show major gaps in certain areas, such as, issues understanding interest rates, inflation, or risk diversification. They state that historically, people relied on government-managed pensions and paid little attention to financial planning which does not apply today anymore. The authors point out the connection between financial illiteracy and riskier behaviors such as late bill payments or excessive interest credit usage. In contrast, financially literate individuals report more planning for retirement and long-term wealth accumulation.

From the behavior side, Atkinson and Messy (2012) note that financial literacy incorporates attitudes and habits that can lead to financial well-being. Their international survey concluded that people with better financial knowledge also display better behaviors including budgeting, saving and timely bill payment. Individuals exposed toward longer-term planning are also likely to adopt healthier behaviors in their financial practices. The research demonstrated notable demographic findings: women scored lower than men on both the knowledge and behavior component, and middle-aged individuals generally outperformed the younger and older age groups. Financial literacy and education were highly correlated, but low-income individuals sometimes showed resistance to outside guidance.

Complementing these findings, Kaye et al. (2014) illustrates that personal finance

behavior is not simply a logical process. Rather, the emotional, culture, and relational contexts of how people manage money influences financial behaviors. Instead of a complete, structured picture of their finances, individuals tend to piece together financial insights through more informal practices, such as only glancing the balance in their account or skimming last week's transactions. This suggests a disconnect between how financial literacy is typically supported and how people actually prefer to engage with their finances. The findings show the importance of designing financial tools that reflect real-life behaviors and habits, and also recognizing that financial decision-making is deeply shaped by individual context and lived experience (Kaye et al. 2014).

Motivation is another aspect to note. Di Domenico et al. (2022) apply Self-Determination Theory framework to reflect on how autonomous, controlled, and amotivated motivation relates to financial knowledge and outcomes. Research demonstrated that individuals who are autonomous, i.e. people who engaged in financial tasks willingly, showed better financial knowledge, self-efficacy, and well-being. While, amotivated individuals, i.e. people lacking interest or feel they cannot manage their finances, were more likely to overspend and reported having lower levels of financial confidence. While both autonomous and controlled motivation were show to correlate with saving and investing, only autonomous motivation had a clear association with positive financial outcomes further mitigating the importance of internal motivation and supportive financial behaviors.

Finally, Lasmiatun (2024) examined how financial education shaped Generation Z's financial awareness and habits. The study found that the understanding and ability of young adults to manage money, strategically plan, and invest was higher for those who participated in the financial education. Although, the understanding of financial competency primarily came from informal learning on social media and things like blogs, instead of formal sources. Lasmiatun (2024) said that participants valued digital content that is presented in an interactive and visual format (infographic, videos, interactive simulations) as they found the smaller chunks of information and visuals easier to engage with and understand. This suggests that improving financial education for Generation Z may require integrating more accessible, digital formats that align with their learning habits and media consumption (Lasmiatun 2024).

2.2 Personal financial planning

Personal financial planning is a lifelong process designed to maximize the use of income and assets, meet financial goals, and reduce uncertainty. Siegel and Yacht (2009) describe this as analogous to planning in other areas: knowing where you are, knowing where you want to go, and figuring out how to get there. They go on to note that there are many interrelated variables and the uncertainty of life can complicate the planning process. Since personal circumstances and the economy change over time, financial planning has to be sufficiently flexible to deal with change while pursuing life-long goals. Financial planning involves continually reassessing goals, considering

alternatives and deciding if changes need to be made (Siegel and Yacht 2009).

In building on this idea, Gitman et al. (2011) note that financial planning is increasingly important given the context of a world in which we cannot expect employer or government support to guarantee long-term security. They describe financial planning as a systematic effort to organize a financial plan for personal goals/values. Their financial planning model has six basic steps: (1) identify and define financial goals (2) develop financial strategies (3) implement financial strategies (4) create and maintain budgets (5) evaluate results using financial statements (6) revise financial plans due to changes in personal circumstances, or unexpected challenges. While success is not guaranteed, consistent and flexible financial planning can improve the likelihood of success in achieving financial security and adapting to the unexpected (Gitman et al. 2011). Huang (2016) states that financial planning is an important skill integrated into daily life, regardless of how much financial considerations affect people in their daily decisions, such as costs being affected by interest rates, increases in annual tuition and others. This is especially important for young people, and college graduates, experiencing independence in an expensive world with low income for the first time. Huang (2016) acknowledged that while some people follow structured paths, and others adopt a more reactive method, financial planning is necessary for eventual success. Similar to Gitman et al. (2011), she also divided the process to six steps: (1) Identifying Smart Financial Goals, (2) Know Your Current Financial Situations, (3) Spending Wisely with a Budget, (4) Managing the Credit, (5) Saving for the Future and (6) Investing the Money.

In the Malaysian context, Lai and Tan (2009) investigated how individuals engage with different areas of financial planning. The results indicated that while budgeting and investment planning is relatively well pursued, financial planning across other areas like insurance, retirement, and estate planning are often disregarded. Specifically, 10% of respondents remarked that they planned across all areas of personal finance. Despite this, there are generally positive attitudes toward budgeting and money, with the conscious recognition of the importance of investment for long term financial security. However, there were less positive attitudes towards retirement, as results indicated greater consideration on health and lifestyle changes. In addition, Lai and Tan (2009) examined job status in financial planning. Findings indicated that employed and self-employed individuals engaged in financial planning more than those without stable income, highlighting that job status has an impact on planning behaviors.

Shah and Bhatt (2014) studied income allocation and investment behavior among residents of the city of Ahmedabad, in the district of Gujarat. Findings indicated that individuals typically allocate 45% of income to consumption, saving 25%, and 15% to investments and emergency funds, with saving priorities of their children's needs, housing, and to save taxes. In addition, respondents expressed that their preferred investments are fixed deposits with banks, insurance plans, and public provident funds. Regarding investment behaviors, male respondents certainly favored investing in stock markets, while female respondents tended to prefer gold, bullion, and PPF investment

products respectively. The above trends may reflect a cautious and security based financial planning behavior focused on meeting long term needs and awareness of investment risk (Shah and Bhatt [2014](#)).

2.3 Personal finance management tools

Personal finance management (PFM) applications have changed from being paper-and-pencil or spreadsheet-based systems, to being advanced web or mobile tools. As Gafrikova et al. ([2015](#)) demonstrate, although using traditional paper-and-pencil, or spreadsheet, methods offers maximized freedom, the downside is greater technical skill and time commitment. Gafrikova et al. ([2015](#)) traced back that dedicated PFM software started appearing in the 1980's, and has since progressed into today's specialized web and mobile tools, many of which are now integrated into banking systems. A key milestone was the launch of Mint.com in 2007, which set a global benchmark for online PFM tools. Despite this, adoption has remained low. Citing a 2012 Aite Group survey they estimate 58% of U.S. consumers have never used and do not plan to use PFM applications, and only 13% of them wanted financial consulting and planning from their financial institutions. In Poland, PFM tools embedded within banks are relatively common but often limited to intra-bank account management, although with higher security standards.

Stefanov et al. ([2024](#)) presented a user-centered design approach for developing a mobile PFM app focusing on the Bulgarian context, where no PFM solutions currently exist. In interviews and survey responses conducted by Stefanov et al. ([2024](#)), they identified practices on how the users wanted to manage their finances: budgeting, monitored expenses, investment planning, and strategy planning. They say that many existing tools fall short due to overly complex interfaces or lack of accessibility. Based on their analysis, they identified some desired functionalities: portable calculator, currency converter, chart extraction, and barcode scanning, all bundled onto a single platform to enhance user experience and effective personal finance management. The significance of design and usability is also verified by Cederberg ([2013](#)), who researched the PFM features young users preferred when embedded in banking apps. She conducted user interviews and prototype testing with their target market while they used the PFM functionalities. Features that the respondents liked included general income and spending diagrams, category-based pie charts, budget tracking with monthly limits, alerts for nearing those limits, and visual planning of fixed expenses. Users identified a desire for easy-to-use applications and preferred minimal manual input with automated categorization features. Users also noted aesthetics were important as they wanted a pleasing and recognizable user interface, as they believed these types of tools would help them more effectively manage their finances due to new insight, control, and goal setting.

Looking at broader range of tools, Torno et al. ([2021](#)) did a taxonomy and cluster analysis on 170 PFM apps, finding ten separate archetypes. The clusters ranged from apps focused on financial news and asset monitoring, to advanced and manual

budgeting tools, to full-featured mobile banking platforms. Most apps required internet access and user accounts and generally, offline functionality was limited to apps that have a basic calculator and manual budgeting possibilities. Very few apps provide an advisory or educational element in their service. Within the clusters, examples such as "Investing with Advice" offer predictive analysis and products while others (e.g., "Transaction Authorisation") are narrower in scope. Despite the wide functionality spectrum, Torno et al. (2021) highlight a lack of educational content across most apps, suggesting a missed opportunity for financial service providers to improve user engagement and trust through better financial literacy tools.

Herrala et al. (2023) explore how PFM systems can both empower and constrain young users, using empathy-based storytelling (MEBS) as a qualitative research method with 191 university students. Participants had significant positive experiences: improved budgeting ability, improved savings and investing practices, and better access to lending and digital transactions, all factors that contribute to personal agency and long-term planning. There were also significant negative experiences: spontaneous spending, difficulty grasping the value of digital money, and over-indebtedness due to ease of credit. Other concerns included stress regarding privacy, surveillance, cybersecurity, overload on information, and over-reliance on technology and digital competence. The study concludes that while PFM tools can promote financial well-being, they may also reinforce behavioral control and cognitive overload when not appropriately designed or supported.

Gamification might offer a partial way around these problems. Bitrián et al. (2021) investigate how gamified features promote user motivation by combining self-determination theory (SDT) with the technology acceptance model (TAM). They evaluated 208 Mint app users and conclude that budgeting, alerts, credit score and tracking opportunities are features that promote competence and autonomy. These feature sets encouraged users' motivation to use the app and resulted in strong perceived usefulness and ease-of-use. Bitrián et al. (2021) suggest independent goal setting, visual display of progress and budgets, real time notifications/alerts, and rewards are elements that app designers can introduce to improve user autonomy and competence. The recommended aspects for both the app designer and user suggest the potential for long-term engagement and a sense of empowerment for the user.

In continuing with a generational perspective, Handayani et al. (2024) described fintech adoption from a millennial perspective since they grew up with technology as a normal way to interact with the world. They examined millennials attitudes toward technology-based financial services and distributed an online survey to 120 users of personal finance apps. The authors felt that personal finance apps helped provide visibility by enabling users to see their income, expenses, and balances in real-time that supported their ability to track their spending behaviour, set their goals, and plan their finances. They presented key features that were seen as motivating factors such as automatic expense tracking, bill reminders, and personalized advice. Concerns about data security and privacy on online platforms did still exist. Fintech applications

have become central to millennials' financial behavior, and their growing usage is pushing traditional financial institutions to adapt and integrate similar technologies to stay competitive (Handayani et al. 2024).

Finally, French et al. (2021) evaluate four PFM apps developed collaboratively with credit unions and tested via a randomized controlled trial with 500 members of Derry Credit Union in Northern Ireland where 260 were in the treatment group and 240 in the control group. The treatment group showed small, significant increases in financial literacy, ability to budget, confidence to budget, and ability to plan for unexpected bills. The treatment group did not show significant increases to digital literacy or behavioural change (i.e., saving money regularly). In terms of use, expectable, descriptive analysis showed that 45% of users used the apps regularly (5 or more times), 32% used the app infrequently, and 23% did not use the app. For the non-user reasons included forgetfulness, relevance, and restrictiveness. Still, most users rated the apps positively, indicating strong potential if future designs can address barriers to usage and engagement.

In summary, the effectiveness of PFM tools lies not only in their technical capabilities but also in their ability to support user behavior, motivation, and emotional well-being. Integrated and user-friendly tools that address privacy concerns, offer personalized insights, and engage users in meaningful way are most likely to foster long-term financial stability and inclusion.

2.4 Design science research

De Sordi (2021) describes Design Science Research Methodology (DSRM) as an approach emerged in the early 2000s to provide a more structured, scientific basis for the development of information systems. Essentially, DSRM aims to solve real-world problems through designing artifacts in a systematic way. Vaishnavi and Kuechler (2015) notes that design is more than simply describing and they define design as purposeful creation of new artifacts, where an artifact can take many forms and types. They say that a design is a routine in a situation when you have enough knowledge to create without the need to develop or extend knowledge, versus a design being labeled innovative when it manifests any form of new knowledge. They continue that the subject of Design Science (DS) is focused on the concepts and processes followed to produce artifacts which comprise constructs, models, methods, and instantiations. This then comprises the practical forms of knowledge needed to solve particular functional needs. Conversely, Design Science Research (DSR) focuses on the scientific study of the design process including analysis, reflection, abstraction, and evaluation. Vaishnavi and Kuechler (2015) explain that DSR contributes to knowledge by investigating and documenting how artifacts have evolved through DSRI and how artifacts achieve their intended use function within contexts.

De Sordi (2021) also describes the difference between natural science, which develops knowledge to examine either physical or abstract phenomena that already

exist, and artificial science, which develops knowledge based in context to develop the creation of human artifacts that aim to solve problems for human populations. According to De Sordi (2021), one defining characteristic of artificial science is a focus on attempting to understand the context of the artifact, in terms of the environment in which will be used or the individual human characteristics and needs that may impact the use of the artifact. As referenced by De Sordi (2021, p. 9), “*Design involves designing something that is a solution to a need, which are called artifacts.*” Artifacts developed using DSRM can take many forms. Vaishnavi and Kuechler (2015) define *constructs*, as the concepts and terms to describe problems and solutions, which may actually evolve in the research. *Models* are organized representation of the relationships among constructs, which may be useful not perfectly accurate. *Methods* are the procedures, algorithms, or techniques used to accomplish desired ends. *Instantiations* are artifact constructs, models, or methods used in practice, which may occur prior to theory being formed (Vaishnavi and Kuechler 2015).

Among the various process models proposed for conducting design science research, the most widely adopted is from Peffers et al. (2007), presented in Figure 1. Their model was developed to address the lack of structure in early DSR efforts and has since become a foundational framework in information systems research. It has a structured six-step process: (1) problem identification and motivation, (2) define objectives for a solution, (3) design and development of the artifact, (4) demonstration of the artifact in use, (5) evaluation of the artifact’s performance, and (6) communication of research results. This structured process will form the methodological basis of this thesis, and is further explained in Chapter 3.

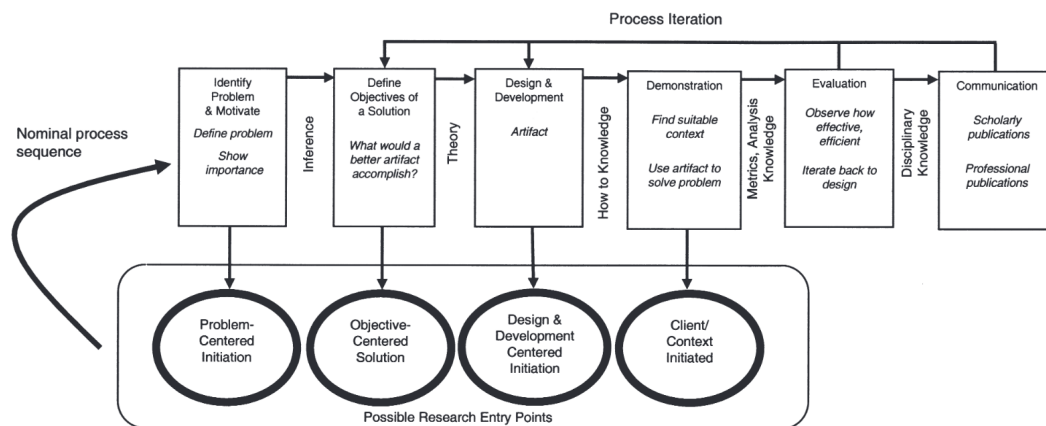


Figure 1: Design Science Process Model (Peffers et al., 2007, p. 54)

In addition to structure and guidelines, DSR also requires an understanding of what a valid contribution means. According to Hevner et al. (2004), design science in information systems requires the replication of artifacts that are not only novel and effective, but also, that have been evaluated objectively in the context of use. The authors provide a balance to the effective usability versus research methodological

rigor by claiming utility as the artifact's valued contribution. In another words, the artifact's reasonable way to address a usable problem. The balance between topical relevance and rigor of method is one of the strengths of design science research, and makes DSRM particularly relevant in applied contexts, such as personal finance technology.

In conclusion, DSRM provide a method to theoretically shape a practical solution to complex problems using iterative artifact development. DSRM encourages including feedback from users, as well as contextual constraints, and formal evaluation. These elements contribute to appropriateness of applying DSRM to develop a more engaging central personal finance application, presented in this thesis.

3 Research methodology

This chapter describes the research methodology used in this thesis, which follows the Design Science Research Methodology (DSRM). The chapter explains how each of the six DSRM phases has been utilized within the scope of this study, from problem identification to communicating results. In addition, the last section provides the justification for why DSRM was selected as the methodology for this thesis.

3.1 DSRM actions

This thesis follows the DSRM proposed by Peffers et al. (2007), which consists of six structured steps for developing and evaluating innovative artifacts. Each step is described in detail below.

3.1.1 Problem identification and motivation

The first step of DSRM, *Problem Identification and Motivation*, focuses on defining a research-worthy problem that holds both academic significance and practical relevance (Peffers et al. 2007). As De Sordi (2021) explains, this requires direct engagement with the field to ensure that the identified challenge reflects real user needs and contextual constraints.

Usability, lack of features, and low demand have all been identified as the most significant limitations facing personal financial management (PFM). Torno et al. (2021) discover that the majority of mobile PFM apps focus on narrow tasks, such as keeping a budget or facilitating payments, and rarely include educational content or consider incorporating holistic financial planning. Herrala et al. (2023) illustrate that while PFM tools can empower users and place them in control of their financial lives, they also run the risk of generating user stress, impulsive behaviors, and create a sense of surveillance.

To validate the problem in this study, a structured questionnaire was conducted with the stakeholder (Table 1), using open-ended questions to investigate challenges and feature gaps. The analysis of the final problem definition is presented in Chapter 4.

Problem identification questionnaire

Q1: How do you currently manage personal finances?

Q2: How often do you use the current solution?

Q3: What different purposes do you use it for?

Q4: What is the most challenging aspect of using the current solution?

Q5: Are there any missing features or functionalities you wish existed?

Q6: How often do you encounter errors, delays, or inconsistencies?

Q7: How would you rate the user experience (UX) and ease of use?

Table 1: Questions used in the problem identification questionnaire

3.1.2 Define the objectives for the solution

The second step of DSRM, *Define Objectives for the Solution*, involves setting goals that respond directly to the identified problem, while ensuring they are feasible and relevant (Peppers et al. 2007). Objectives can be both quantitative, such as measurable improvements, and qualitative, such as addressing user frustrations or unmet needs. As De Sordi (2021) notes, this stage includes assessing the current state of the problem, reviewing existing tools, and identifying how the proposed artifact can offer a suitable solution.

To formalize the solution scope, the system requirements were divided into functional and non-functional categories. Functional requirements define what the system should do, such as authentication, data processing, and reporting, while non-functional requirements specify how the system should perform, including its reliability, security, and responsiveness (Ironhack 2024). This structure helps ensure the artifact meets both usability expectations and technical standards.

To focus development on the most critical features, a prioritization of the requirements was conducted with the stakeholder. The MoSCoW method (Brush 2023), a common approach in Agile development, was adapted to suit the project context. The customized scale, shown in Table 2, includes four levels: Must have, Should have, Nice to have, and Optional.

Prioritization	Description
Must have	Mandatory in the final product
Should have	Essential but not absolutely vital
Nice to have	Useful additions, but not necessary
Optional	Could be added later

Table 2: Custom prioritization scale for the context case

This customized ranking allowed the stakeholder to distinguish core features from those better suited for future iterations. Requirements were labeled as functional requirements (FR) or quality attributes (QA), and numbered by priority, with lower numbers indicating higher importance. The requirements are presented in detail in Chapter 5.

3.1.3 Design and development

The third step of DSRM, *Design and Development*, involves specifying the system architecture and core functionality, followed by the implementation of the artifact (Peppers et al. 2007). According to De Sordi (2021), this phase is iterative and includes ongoing development and evaluation to ensure that the artifact effectively addresses the defined problem. Continuous refinement based on stakeholder feedback is encouraged to enhance both functionality and relevance.

Architectural decisions were made based on the prioritized requirements. Enstrom (2018) defines Architectural Decisions (ADs) as significant, documented choices that influence the system's structure, including software technologies, data integration, and maintainability strategies. These decisions are distinct from routine design details and serve to align the system with long-term goals and constraints (Enstrom 2018). The key ADs for this project are presented in Chapter 6.

After defining the architecture, the artifact was developed in iterative cycles. Feedback from the stakeholder was used to refine features, user workflows, and interface elements. This is motivated from the findings from literature as both Stefanov et al. (2024) and Cederberg (2013) noted the importance of usability, simplified and inviting interface, and centralized functionality.

3.1.4 Demonstration

The fourth step of DSRM, *Demonstration*, involves showing that the developed artifact can solve at least one instance of the identified problem (Peffers et al. 2007). This may involve experimentation, simulations, case studies, or other forms of proof. As described by De Sordi (2021), this phase aims to validate whether the artifact's structure, procedures, and outputs address the problem in a meaningful way, even if full performance optimization has not yet been achieved. De Sordi (2021) also notes that demonstration is sometimes incorrectly reported as evaluation in academic studies, especially when the artifact is still in development.

In this thesis, demonstration was used to show how the artifact addresses key issues identified during stakeholder analysis. These included fragmented tools, repetitive manual updates, and limited trust in the accuracy of financial data. These same challenges were also noted in the literature. Gafrikova et al. (2015) mentions that spreadsheet-based solutions require time and technical skills. Cederberg (2013) found that users might stop using the tool if it requires too much manual input, encouraging automation. Additionally, Herrala et al. (2023) found that financial tools perceived as unreliable or difficult to use may increase user stress and undermine their sense of control.

The demonstration presents the following capabilities of the system:

- Addressing complexity and reducing human error
- Solving scalability limitations
- Automating financial data retrieval
- Providing centralized tracking across financial categories
- Delivering integrated and efficient reporting

3.1.5 Evaluation

The fifth step of DSRM, *Evaluation*, assesses how well the artifact solves the identified problem by comparing its intended objectives with the actual outcomes (Peppers et al. 2007). This step involves analyzing the artifact's performance, usability, and functional alignment with user needs. Evaluation is important for determining whether the artifact works as intended in reality. De Sordi (2021) explain that there are artificial evaluations occurring under isolated or theoretical conditions, and naturalistic evaluations that occur within real environments, with real users, and real problems. Naturalistic evaluations are aligned with the fundamental principles of design science, particularly in applied domains.

In this study, the artifact was evaluated using four different methods. First, task-based user testing was conducted to assess usability and alignment with expected workflows. The stakeholder performed typical usage scenarios independently on their own device while being observed. This allowed for direct feedback on task completion, design effectiveness, and interface clarity. The scenarios are listed in Table 3.

User testing tasks
T1: Add a new transaction for every asset category and review updated net worth
T2: View and interpret asset distribution using pie charts and asset tables
T3: Add an expense and income and review the updated cash flow visualization
T4: Generate a financial report for the past three months

Table 3: Task-based user testing

Following the test session, the stakeholder completed a structured evaluation questionnaire in similar format as in problem identification. The questions were grouped into four categories: general impressions, feature satisfaction, problem-solving, and suggestions for improvement (Table 4).

Evaluation questionnaire
General impressions
How intuitive did you find the application interface?
Was it easy to find and use the key features you needed?
Feature satisfaction
How satisfied are you with the asset tracking functionality?
How satisfied are you with the cash flow tracking functionality?
Do you feel the visualizations (charts, tables) are helpful and informative?
How useful is the PDF report generation compared to your previous manual process?
Problem solving
To what extent has this solution reduced manual work compared to your old system?
Do you feel this solution helps you avoid errors or inconsistencies?
Has your confidence in your financial data improved?
Suggestions and improvements
Is there anything missing you would expect in a full solution?
What would you improve in future versions?

Table 4: Questions used in the evaluation questionnaire

In addition to collecting qualitative feedback, the evaluation included a comparison with the stakeholder’s baseline system. This helped identify gains in automation, usability, and reporting quality. Finally, an analytical review was performed by checking how well the implemented solution satisfied the original design objectives from earlier DSRM stages. The evaluation results are presented in Chapter 7.

3.1.6 Communication

The sixth and final step of DSRM, *Communication*, focuses on sharing the research outcomes with both academic and professional audiences. According to Peffers et al. (2007, p. 92), “*Communicate the problem and its importance, the artifact, its utility and novelty, the rigor of its design, and its effectiveness to researchers and other relevant audiences, such as practicing professionals, when appropriate.*” De Sordi (2021) highlights that effective communication in DSR must be tailored to its audience. He says that for practitioners, the emphasis is on the artifact’s purpose and practical utility. For researchers, the focus is on presenting the design process, structure, and evaluation outcomes to ensure methodological rigor and reproducibility (De Sordi 2021).

This thesis represents the communication phase of the DSRM process. It documents all stages of the research, beginning with problem identification and continuing through design, development, demonstration, and evaluation. The aim is to present the motivation for the work, the rigor of its execution, and its contribution to both

research and practice. By clearly articulating the methodology and results, this thesis contributes to ongoing discussions about the design of personal finance tools and the practical application of DSRM to real-world challenges.

3.2 Motivation for DSRM

DSRM is well suited for this thesis due to its structured and iterative process for developing and evaluating artifacts that address real-world problems. This study aims to create a proof-of-concept (PoC) of personal finance management application that fulfills a stakeholder's specific needs by using automation and data integration to provide better financial insight. DSRM provides an organized and systematic approach as an avenue to develop, validate and refine such an artifact ensuring its effectiveness and relevance in practice. This was also pointed by Torno et al. (2021) who stated that design science research is a promising practice to develop innovative IT artifacts.

The methodology is specifically intended for research where the goal is to construct and evaluate artifacts that extend existing solutions or introduce new functionality (Peppers et al. 2007). As Hevner et al. (2004) explains, design science research is most effective when artifacts are developed in response to clearly defined business needs and evaluated within real-world contexts. This aligns with the aims of this thesis, which addresses limitations in current financial tracking tools such as manual spreadsheets that lack scalability, automation, and integration. Through DSRM, these limitations can be systematically addressed using an iterative development process. The six-phase structure provides a transparent development roadmap, making it easier to track progress and validate design decisions.

DSRM differs from purely theoretical approaches by including demonstration and practical evaluation. Because this project involves a real stakeholder with specific financial tracking needs, the methodology supports continuous engagement to ensure that the resulting solution meets actual user expectations. As noted by De Sordi (2021), practitioner involvement strengthens both academic rigor and real-world applicability. The evaluation phase allows the artifact to be assessed against predefined objectives and revised if necessary. Since the goal of the thesis is to develop a PoC rather than a production system, DSRM also allows for design flexibility and exploration of feasibility without requiring full deployment. In particular, the Demonstration phase enables targeted testing of core functionality in a controlled setting, helping confirm the solution's effectiveness before further development (Peppers et al. 2007; Hevner et al. 2004).

4 Problem identification and motivation

This chapter presents the problem description that the proof-of-concept application aims to address. It begins with insights gathered from a structured questionnaire conducted with the stakeholder, followed by a detailed description of the current financial management approach and its associated challenges. Finally, the chapter explains the motivation for developing a new solution, based on the limitations and problems identified in the existing system.

4.1 Stakeholder insights from questionnaire

To gain a deeper understanding of the problem, a structured questionnaire was conducted with the stakeholder, as introduced in Chapter 3.1.1. Several challenges were found from the financial management approach based on the responses.

4.1.1 Current financial management approach

The stakeholder uses multiple financial tools to track income, expenses, and investments. Their process includes:

- Tools used: A combination of Spendee (for expense tracking), multiple bank accounts across two countries, Google Sheets (for share price and exchange rate tracking), and several Excel spreadsheets for real estate and mortgage tracking.
- Manual data entry: Financial data is periodically fetched from various sources and consolidated into a “master” Excel file, in use since 2014.
- Financial reporting: Quarterly and yearly reports are generated manually by analyzing data and producing graphs.

4.1.2 Frequency of use

The stakeholder engages with the financial system at varying intervals:

- Daily: Tracking expenses in Spendee.
- Weekly: Updating and maintaining Excel files.
- Quarterly: Conducting a full financial review and generating reports.

4.1.3 Challenges with the current system

The stakeholder identified the following major issues:

- Complexity and human error: The system has become difficult to maintain and prone to mistakes.

- Scalability limitations: Excel structures struggle to handle growing financial data needs.
- Lack of automation: No integration with online sources and updates must be done manually.
- Reporting inefficiencies: The quarterly report process is time-consuming and fragmented.

4.1.4 Desired features for an improved solution

The stakeholder expressed the need for:

- More automation: A system that tracks key metrics with minimal manual effort.
- Better data integration: Automatic synchronization with sources like stock prices, exchange rates, and banks.
- A centralized platform: A single system combining all financial aspects.
- Enhanced accuracy: Reduced human error and more reliable data.

4.1.5 Stakeholder reflections

The stakeholder summarized their concerns as follows: *"The whole process is getting overly complex, resulting in a higher probability of human error impacting data quality. Any changes made to the document might break something due to complex dependencies across different data tabs."*

Although they find Excel somewhat enjoyable and familiar, they acknowledge the system is not user-friendly for others and is increasingly difficult to manage. As they put it: *"For me, it is somewhat of a relaxing experience, but it is starting to feel overwhelming because of the inaccuracies and gaps that hinder me from getting accurate data."*

4.2 Problem description

The stakeholder has tracked income, expenses, and investments for over a decade using a combination of spreadsheets, financial applications, and manual data collection. Over time, this system evolved to support detailed financial analysis, including quarterly and yearly reporting. However, as financial data has grown in volume and complexity, maintaining the system has become increasingly difficult.

The current setup involves multiple tools: Spendee for expense tracking, bank accounts across two countries, and Excel spreadsheets for managing real estate investments, mortgages, and stock portfolios. These fragmented sources require

manual consolidation into a “master” Excel file. Data must be periodically fetched, entered, and processed by hand, resulting in a time-consuming workflow that is increasingly prone to error due to complex interdependencies across formulas and sheets.

The main challenges of the existing system include:

- **Scalability limitations** – Excel struggles to accommodate the increasing volume and complexity of financial data.
- **High risk of error** – Manual updates to stock prices, exchange rates, and balances introduce inconsistencies and inaccuracies.
- **Lack of automation** – No integration with online financial data sources requires frequent copy-paste operations and manual data entry.
- **Inefficient reporting** – Quarterly reporting involves multiple manual steps: collecting data, updating formulas, generating charts, and compiling summaries.

The stakeholder is very sophisticated in their financial management, but as their system complexity and fragmentation grow, inefficiencies and the risk of error has increased. As many of the authors note, working with disparate tools or manual processes creates usability issues, deters engagement, and leads to errors in financial decision-making (Gafrikova et al. 2015; Torno et al. 2021; Stefanov et al. 2024). Research also shows that even knowledgeable users cannot maintain their desired financial behaviours (or make decisions) when the available tools do not simplify their decision-making or provide timely feedback related to their current financial goals (Atkinson and Messy 2012; Kaye et al. 2014; Cederberg 2013). This behaviour supports the concepts of integrated ease of use, automation, real-time data, and supporting feedback loops toward the establishment of habits, all of which were also requested by user participants in the most recent fintech and PFM adoption studies (Bitrián et al. 2021; Handayani et al. 2024)

4.3 Motivation

The stakeholder’s difficulties represent a wider demand for personal finance tools that combine automation, accuracy, and usability. Their current system allows them to track all aspects of their finances in great detail, though, it is becoming more and more difficult to sustain, as well as requiring scaling. The stakeholder has not considered commercial tools, because the stakeholder believe their needs to be really specific, they would be unable to find an off-the-shelf product that fulfilled their needs. It is noted by Stefanov et al. (2024), that the absence of localized tools (e.g. applications in Bulgarian) could also lead to unmet expectations for users in certain regions. Torno et al. (2021) also note that most personal finance applications tend to focus on a single function (budgeting versus payments) than being a solution for everything, and as such, users often need to resort to some other method, for example use of spreadsheets.

Besides missing features, many existing tools do not offer a complete solution and force users to manage their finances across several separate apps. While automation was recognized as a major priority, the stakeholder also indicated that accuracy and completeness are critical aspects of their financial data. They said that without these, they may not justify the expense of moving to another system in the future. This viewpoint is aligned with Cederberg (2013), who found that users would often abandon tools that required excessive manual effort even if they provided some level of automation or tracking.

An improved solution should address the following key needs:

- Automated financial data retrieval to eliminate repetitive manual updates.
- Centralized tracking of financial data across multiple accounts and investment platforms.
- Integrated analytics and reporting to generate insights with minimal user intervention.
- Enhanced accuracy and reliability to reduce the risk of calculation errors.

The motivation for this research is based on creating a scalable personal finance system that can improve data reliability, reduce manual effort, and allow individuals to make informed decisions. These priorities match the recommendations in the literature for creating tools that not only automate and do financial tasks, but can motivate users, give feedback, and support the financial goals of the user (Bitrián et al. 2021; Shafiee et al. 2024). While some studies also mention potential stress or overreliance on digital systems, they note the importance of user-centered and emotionally supportive design (Herrala et al. 2023).

5 Defining objectives for the solution

This chapter defines the objectives of the proposed solution based on the problem analysis and stakeholder needs presented in Chapter 4. It presents both functional and non-functional requirements that guide the development of the proof-of-concept application. These requirements are then prioritized to ensure that the most critical features are addressed.

5.1 Requirements

The following requirements are intended to resolve the major problems and motivations explained in Chapter 4. These requirements reflect user needs and incorporate some of best practices in software development and PFM design. The requirements are related to practices mentioned in the literature where fragmented and manual systems have been shown to be inefficient and create cognitive burden and low usability (Gafrikova et al. 2015; Herrala et al. 2023). In contrast, research also highlights that effective personal finance tools build long-term engagement by providing automated features, availability, ease of use, and tools to create engagement and maintain motivation (Bitrián et al. 2021; Cederberg 2013).

The requirements are divided into two main categories: functional and non-functional. Tables 5 and 6 present a summary of each, including descriptions, user stories, and acceptance criteria that define when the requirement is considered fulfilled.

The functional requirements focus on features that support an understanding of financial visibility, decision making, and operational efficiency. For example, dashboards, tags, and reporting features. These features were requested by the stakeholder, and research also suggests that clear goal-oriented feedback increases engagement and personal oversight (Shafiee et al. 2024).

Functional requirement	User story	Acceptance
Present real-time capital information	"As a user, I want to see the real-time value of all my assets."	The system fetches up-to-date stock prices and displays the real-time value of all tracked assets.
User interface (Desktop and mobile)	"As a user, I want to be able to add/update/view my financial data from mobile and desktop."	The application provides a responsive user interface that works on both desktop and mobile devices.
Data visualization of capital and statistics	"As a user, I want to see how my assets are growing in a way that supports me in making financial decisions."	The application displays interactive visualizations, such as charts and tables, representing asset growth and financial trends.

Functional requirement	User story	Acceptance
Manual financial data input	"As a user, I want to be able to add/modify/delete income, expenses, and different assets through the application or import them from a file."	Users can manually input and manage financial records and import data via supported file formats.
Automatic data fetching	"As a user, I want my income, expenses, transactions, and different assets to be fetched automatically from the bank and brokers that I use."	The system integrates with external APIs or services to automatically retrieve user financial data.
Tagging	"As a user, I want to be able to divide my assets into different categories and subcategories."	Users can assign, edit, and view custom categories and subcategories for each asset.
Financial report generation	"As a user, I want to receive on-demand financial reports (weekly/monthly) summarizing my income, expenses, and asset growth."	Users can generate and view downloadable reports summarizing key financial metrics over a selected time range.
Customizable dashboards	"As a user, I want to customize the widgets on my dashboard to suit my financial goals"	Users can add, remove, and rearrange dashboard elements such as charts, summaries, and alerts.
Choose currency	"As a user, I want to be able to choose the currency I want to use for data visualization and analytics."	Users can select a preferred display currency, and the system automatically converts values using current exchange rates.

Table 5: Functional requirements

The non-functional requirements address factors such as trust, reliability, and responsiveness which are important factors for user adoption and continued engagement. Herrala et al. (2023) noted that technical issues, user errors and concerns about data security resulted in increased stress and lower rates of user engagement. Handayani et al. (2024) identified privacy and information security as major concerns for millennial users.

Category	Non-Functional requirement	User story	Acceptance
Data protection and security	Prevent unauthorized parties from accessing data.	“As a user, I want my personal data to be safe and not get into the wrong hands.” “As an application owner, I want to protect customer and business data from unauthorized access.”	All user data is isolated and accessible only to the authenticated user. External access is restricted via secure authentication and authorization mechanisms.
Reliability	The statistics provided by the system should be correct.	“As a user, I need to trust the calculation and analytics given by the system.”	The system produces correct financial analytics with an accuracy rate of at least 99.99% when validated against sample data.
Performance	Good response time	“As a user, I want the system to process my actions, such as updating assets and fetching reports, quickly.”	95% of user requests are processed within 1 second under normal operating conditions.

Table 6: Non-Functional requirements

5.2 Prioritized objectives

After the requirements for the proof-of-concept application were defined, they were prioritized together with the stakeholder, as described in Section 3.1.2. The prioritized list is presented in Table 7, including a short description, associated goal, and priority level for each requirement.

Label	Description	Goal	Priority
FR-01	User interface (Desktop and mobile)	Ensure the application has a responsive and intuitive interface usable across devices.	Must have
FR-02	Data visualization of capital and statistics	Provide interactive and clear visualizations to help users analyze financial performance.	Must have

Label	Description	Goal	Priority
FR-03	Tagging	Allow users to organize financial data using custom categories and tags.	Must have
QA-01	The statistics provided by the system must be correct	Ensure that all system-generated statistics and analytics are accurate and reliable.	Must have
FR-04	Present real-time capital information	Display up-to-date values of all user assets using real-time data sources.	Should have
FR-05	Financial report generation	Generate reports that summarize financial metrics for selected periods.	Should have
QA-02	Prevent unauthorized parties from accessing data	Secure all user data and prevent unauthorized access through authentication and encryption.	Should have
FR-06	Automatic data fetching	Integrate with financial APIs to import transaction and asset data automatically.	Nice to have
FR-07	Choose currency	Allow users to choose their preferred display currency and apply real-time conversion.	Nice to have
FR-08	Customizable dashboards	Let users tailor their dashboards with widgets for personalized financial tracking.	Nice to have
FR-09	Manual financial data input	Support manual entry for transactions and assets when automation is unavailable.	Optional
QA-03	Good response time	Ensure fast and responsive interaction for common user actions.	Optional

Table 7: Prioritized requirements

6 Design and development

The chapter provides the technical basis for the proof-of-concept application. The chapter presents the major design decisions, system architecture, and implementation details that were used to meet the functional and non-functional requirements described in the previous chapter. The focus was on designing a modular, scalable and maintainable system that met stakeholder requirements while facilitating rapid development under the scope of a PoC.

6.1 Architectural decisions

Based on the prioritized functional requirements (FR) and quality attributes (QA) presented in Chapter 5, key architectural decisions were made to support the most critical goals for the system. The focus was placed on addressing the “Must have” and “Should have” requirements. Since this is a proof-of-concept (PoC) application, the architecture is intentionally simplified to focus feasibility, core functionality, and rapid development while minimizing unnecessary complexity. The architectural decisions are presented in Table 8, along with their justifications and corresponding requirement links.

The technical design reflects both stakeholder needs and recommendations from the literature on PFM tools. Previous studies indicated the importance of centralization and modularity for facilitating sustained usage and continued engagement (Gafrikova et al. 2015; Torno et al. 2021). Similarly, usability and reliability were important factors. Cederberg (2013) noted the need for an intuitive user interface and automated categorization to minimize user effort, while Herrala et al. (2023) recognized that technical complexity and user error may lead to stress and lower user trust. The decision to create a cross platform UI, and modular back end is meant to promote flexibility and scalability. Additionally, features like visual analytics and automation were chosen based on findings that they enhance user motivation and promote clearer financial understanding (Shafiee et al. 2024; Cederberg 2013).

Short description	Decision	Justification	Requirement
User interface	- Use Flutter for a cross-platform UI (mobile and desktop). - Implement a basic responsive UI.	- Flutter allows quick UI iteration with a single code base. - Responsive UI ensures usability across devices.	FR-01

Short description	Decision	Justification	Requirement Link
Backend services	- Use ASP.NET Core Web API for backend logic (C#). - Implement RESTful API for data flow. - Use MongoDB.Driver for data access. - Separate frontend and backend into modular components.	- ASP.NET Core is performant, scalable, and aligns with developer expertise. - Using MongoDB.Driver ensures compatibility and flexibility with MongoDB data models. - Modular architecture supports maintainability and future scaling.	FR-01, QA-02
Database	- Use MongoDB for storing financial data, tags, and transactions. - Define collections for structured data storage.	- Flexible schema allows adapting to evolving financial data.	FR-03, FR-06, QA-01
Data visualization	- Use chart libraries and built in tables in Flutter for financial data visualization.	- Charts and tables help users analyze financial trends effectively.	FR-02
Automatic data fetching	- Use Google Sheet API to fetch share prices. - Implement scheduled jobs to fetch and store data.	- Google Sheet integration ensures real-time financial updates with no costs. - Scheduled jobs automate data retrieval.	FR-04, FR-06
Multi-currency support	- Store all financial data in a base currency (e.g., USD/EUR). - Fetch exchange rates from HexaRate API.	- Storing data in a base currency avoids unnecessary conversions. - Dynamic conversion provides user flexibility.	FR-07
Customizable dashboards	- Store dashboard layouts in MongoDB as JSON documents. - Use Flutter's Drag-and-Drop Widgets.	- JSON storage enables flexible UI customization. - Drag-and-Drop UI enhances usability.	FR-08
Manual financial data input	- Use different forms to input different transactions manually. - Make tables editable. - Support CSV imports.	- Enables easy data input when integrations might not be available. - CSV import reduces manual workload for bulk data entry.	FR-09

Short description	Decision	Justification	Requirement Link
Performance optimization	- Optimize API and DB queries with indexing and caching. - Use providers in UI and only call API when something is updated.	- Reduces response times and improves scalability.	QA-03

Table 8: Architectural decisions and their linkage to requirements

6.2 System architecture

The high-level system architecture is shown in Figure 2. The system consists of three core components aligned with the architectural decisions discussed in the previous section: the user interface, the backend API, and the data storage layer. Additionally the two integrations are presented in the picture: stock prices and exchange rates.

The user interface is developed using Flutter and Dart to support cross-platform deployment across mobile and web platforms. It handles user interactions, such as adding or editing transactions, and visualizes financial data through interactive charts and tables. While WinUI was initially considered for the front end, it was eventually let go due to its limited visualization capabilities and its restriction to the Windows desktop environment.

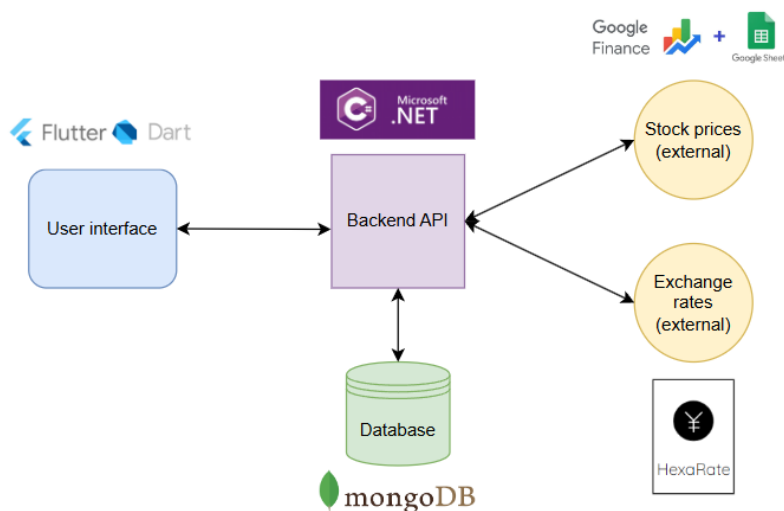


Figure 2: High-level system architecture diagram

The backend API is implemented in C# using the .NET framework. It contains the core business logic of the application, including transaction handling (shares,

cash, and real estate), statistical computations, and communication with external data sources. The API is also responsible for all interactions with the database, including data storage, updates, and retrieval. While this design does not scale far in production environment where there is more traffic, the code is structured in a way that would allow separation to microservices later on.

MongoDB is used as the NoSQL database due to its flexibility in managing diverse financial data. It stores all the data models, including transactions, asset details (shares, real estate, and cash), cash flow (income and expenses), and daily net worth snapshots.

To enrich the application with current market data, two external integrations are included. For stock prices, the system uses a Google Sheets integration powered by Google Finance. Real-time prices are retrieved via the Google Sheets API based on ticker symbols, with data refreshed each time the application is opened or modified. Although this solution requires some manual setup by users, it was selected due to the high cost and limited availability of commercial stock APIs.

Currency exchange rates are fetched from the HexaRate API. To improve efficiency, exchange rates are retrieved once per day and cached in the database for reuse, ensuring up-to-date conversion data with minimal external requests.

6.3 Implementation details

6.3.1 Data model and domain objects

The application's data model is designed to reflect the financial domain and support core functionalities such as asset tracking, income and expense monitoring, and net worth calculation. A NoSQL, document-based approach is used, with each domain object stored in a separate MongoDB collection. The main collections and their roles are described below:

- **ShareTransaction, CashTransaction, RealEstateTransaction:** These collections store raw transaction data entered by users. Each record includes metadata such as transaction date, amount, and relevant domain-specific fields such as purchase price or loan amount for real estate. These transactions form the base for calculating asset values.
- **Share, Cash, RealEstate:** These collections represent the user's current asset state. They are computed dynamically from transaction history. For example, the Share collection aggregates holdings per ticker symbol, calculating total quantity and current market value.
- **CashFlowTransaction:** This collection tracks income and expense activity. Each entry includes the transaction type (income or expense), category, amount, currency, and optional notes. This enables budget tracking, category breakdowns, and monthly expense summaries.

- **NetWorthSnapShot:** This collection stores daily snapshots of the user's total net worth. Whenever assets are modified, for example, through a new or updated transaction, the system either creates or updates the snapshot for that date. These records support historical net worth tracking and trend visualization.
- **ExchangeRate:** This collection stores daily currency exchange rates retrieved from the HexaRate API. Rates are fetched once per day and cached to reduce external requests. All financial data are normalized to a base currency (e.g., EUR or USD) using these stored rates.

This modular data model supports a flexible and extensible architecture, enabling new features to be added with minimal disruption to existing components. The backend API follows a structured design pattern in which each domain feature is managed through a dedicated `Controller–Logic` pair, promoting a clear separation between request handling and business logic. This pattern also enables possible later separation to their own microservices to increase scalability.

6.3.2 User interface structure and design approach

The user interface is implemented in Flutter using a modular, screen-based architecture. Each primary functionality, such as the dashboard, asset management, transaction input, and cash flow tracking, is encapsulated in a dedicated screen. These screens are composed of reusable UI components, including editable tables, input forms, pie charts, and bar graphs. All components are located in the `widgets/` folder and access application data through state management provided by the provider package. The interface updates reactively in response to changes in the underlying data layer, ensuring synchronization between the data model and the rendered views.

Financial visualizations are built using the `fl_chart` library, supporting different diagrams such as line, bar, and pie charts. These provide users with immediate visual feedback on financial distributions and trends. As noted by Bitrián et al. (2021), visual progress indicators and dashboards can improve user motivation and purposefulness. To support responsive layouts, Flutter's layout tools, such as `Expanded`, `Flexible`, and `LayoutBuilder`, are used alongside maximum width constraints to optimize the interface for large screens. While the current design primarily targets desktop and web resolutions, the use of Flutter enables future mobile support with minimal code changes.

The UI design focuses clarity and ease of data entry. Structured forms are provided for each transaction type, and editable tables enable efficient bulk modifications. Usability and data integrity are further improved through features such as real-time form validation, dropdown menus, and multi-select checkboxes. This emphasis on visual feedback and structured input aligns with design recommendations for PFM tools, which stress that reducing manual effort and simplifying interactions encourages continued engagement and reduces user stress. Cederberg (2013) highlights that young

users value tools that have inviting interface, automated categorization, and minimized input complexity, while Herrala et al. (2023) caution that systems perceived as difficult or unreliable can increase cognitive load and undermine trust.

6.3.3 Development tools and process

A range of modern tools were utilized during the PoC development. Backend services were developed using Microsoft Visual Studio Community 2022, while the Flutter-based frontend was implemented in Visual Studio Code. MongoDB Compass was used to inspect and manage database documents during development. Version control was handled using Git, enabling structured branching, change tracking, and safe iteration.

Generative AI tools were used to support the development process by assisting with code scaffolding, debugging suggestions, and exploration of architectural patterns. These tools were particularly helpful in accelerating UI layout design in Flutter, formulating MongoDB queries, and structuring backend logic in .NET. However, their role was strictly supportive: all code was manually reviewed, adapted, and integrated to meet the project's specific requirements.

An iterative development approach was followed, including regular feedback from the stakeholder. The functionality was implemented in modular units. For example, share management and its related transactions were treated as a single feature set. Each feature typically followed a consistent workflow: beginning with UI design, followed by data contract definition, backend logic and API endpoint creation, database integration, and finally frontend-backend connection. After each feature reached a working state, time was allocated for refactoring and fine-tuning before proceeding to the next set of features. This incremental approach supported rapid prototyping while allowing flexibility to respond to evolving requirements throughout the development phase. Iterative development with stakeholder feedback is also recommended in DSR literature. Hevner et al. (2004) stated that iteration and implementation are central pieces to this research as it increases the quality of the end product and helps address problems along the way.

7 Demonstration

This chapter describes a practical demonstration of the proof-of-concept application. The goal is to demonstrate how the implemented system solves the main problems in the stakeholder's financial management process. The demonstration will follow a structure of the most important issues and how the proposed solution addresses these.

7.1 Addressing complexity and human error

A major challenge with the stakeholder's current system is its complexity and vulnerability to human error, caused by the use of multiple spreadsheets and manual calculations across various financial sources. The PoC application mitigates these issues by consolidating all financial data into a single, structured backend supported by well-defined logic and calculations.

Each financial domain (shares, cash, and real estate) is handled by a dedicated logic class in the backend, following a consistent Controller–Logic design pattern. This separation ensures that each logic class is responsible for domain-specific operations such as asset aggregation, transaction handling, and profit calculation. For example, the `ShareLogic` class compiles a user's holdings based on transaction history and computes key metrics, including:

- Total invested amount
- Current market value (based on the latest prices)
- Profit or loss in absolute terms
- Profit or loss as a percentage

These calculations are performed automatically on the server side, using the user's base currency and up-to-date exchange rates and share prices. The results are returned to the frontend as part of the asset model, eliminating the need for spreadsheet-based formulas or manual input. Similar logic applies to cash and real estate assets, with real estate computations taking into account the purchase price and loan amount.

By encapsulating business logic in code, the system delivers consistent and reliable results, significantly reducing the risk of errors caused by user input, broken formulas, or accidental modifications. Centralized logic also simplifies future updates as any change to calculation rules, such as profit formulas or exchange rate handling, can be applied system-wide from a single source.

The frontend presents these results visually and interactively. On the Dashboard page, shares are displayed in a doughnut chart (Figure 3), while the Assets and Transactions page shows detailed asset data in editable tables (Figure 4).

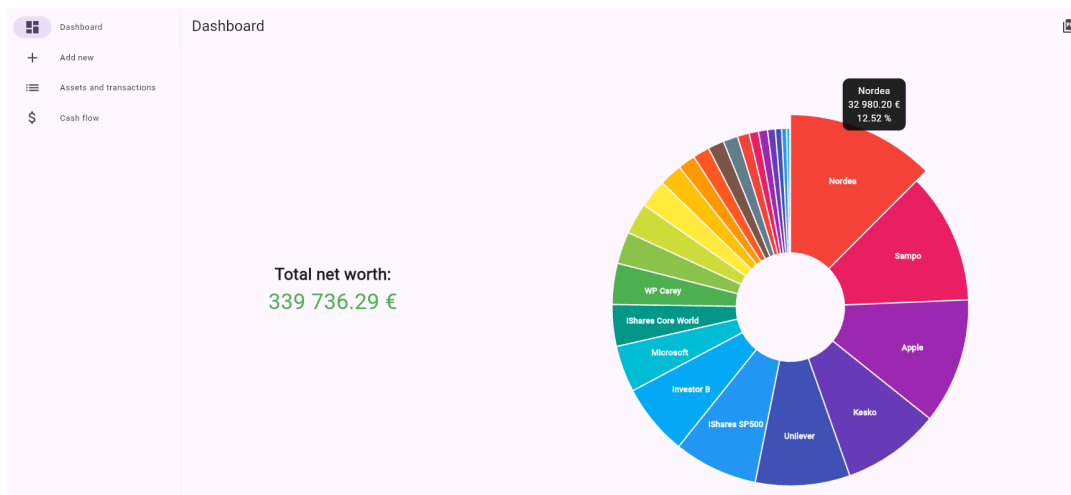


Figure 3: User's shares presented in an interactive doughnut chart

Dashboard

+ Add new

Assets and transactions

\$ Cash flow

My assets and transactions

Assets

Transactions

Shares

Edit

Name	Value	Percentage	Current price	Amount	Country	Average price	Currency	Profit (€)	Profit (%)	Category	Sub category
Nordea	33 111.70 €	12.56 %	12.59 €	2630	EU	8.05 €	EUR	13 862.96 €	72.02 %	Dividend	Yield
Sampo	31 348.00 €	11.89 %	9.22 €	3400	EU	6.97 €	EUR	9 594.40 €	44.10 %	Dividend	Yield
Apple	29 716.36 €	11.27 %	208.53 €	160	US	103.67 €	USD	20 002.74 €	205.92 %	Growth	Value
Kesko	23 385.60 €	8.87 %	20.16 €	1160	EU	18.67 €	EUR	4 863.95 €	26.26 %	Dividend	Yield
Unilever	22 664.80 €	8.60 %	55.28 €	410	EU	46.29 €	EUR	4 210.87 €	22.82 %	Dividend	Yield
iShares SP500	20 016.00 €	7.59 %	556.00 €	36	US	568.04 €	EUR	- 802.27 €	-3.85 %	Growth	ETF
Investor B	17 145.72 €	6.50 %	294.65 €	635	EU	242.75 €	SEK	6 817.12 €	66.00 %	Growth	Value
Microsoft	11 056.35 €	4.19 %	443.35 €	28	US	307.65 €	USD	2 885.63 €	35.32 %	Growth	Value
iShares Core World	9 942.00 €	3.77 %	99.42 €	100	US	103.66 €	EUR	- 411.91 €	-3.98 %	Growth	ETF
WP Carey	9 801.78 €	3.72 %	61.14 €	180	US	65.54 €	USD	- 541.47 €	-5.23 %	Dividend	REIT
Total value: 263 646.81 €											

Cash

Edit

Account name	Amount	Currency	Value as EUR
Säästötili	70000.00 SEK	SEK	6 414.66 €
Total value: 6 414.66 €			

Real estate

Edit

Date bought	Property name	Purchase price	Loan amount	Ownership value
06.01.2021	Lärkaninpolku	200 000.00 €	150 000.00 €	50 000.00 €
09.01.2022	Tullilintu	250 000.00 €	230 000.00 €	20 000.00 €
Total value: 70 000.00 €				
Total debt: 380 000.00 €				

Figure 4: User's assets presented in editable tables with summary statistics

Automating complex calculations and reducing manual workflows increases ease of use. This is consistent with research that suggests users are more likely to engage with systems with low effort and clear feedback. Cederberg (2013) found that young users preferred financial tools that required less manual input, included automated categorization, and improved visual clarity which further empowered them to manage their finances better and encouraged continued use. Likewise, Herrala et al. (2023) identified that by making financial tasks easier to complete, user stress was reduced, and sense of control increased.

7.2 Solving scalability limitations

The stakeholder's previous system, based on Excel spreadsheets, had scalability limitations as the financial data volume increased. Over time, the growing number of transactions, investment types, and calculated fields made the system fragile and difficult to scale. Complex dependencies between formulas and worksheets created a risk where even minor changes could disrupt the entire setup. The PoC application addresses these challenges through a service-oriented architecture and a document-based database model.

Financial data is stored in MongoDB collections, which are convenient for evolving data structures. This enables the system to handle increasing volumes of transactions and assets without requiring strict schemas or manual reconfiguration. The backend is implemented as a stateless ASP.NET Core Web API, which supports horizontal scalability. While this PoC is not intended for production deployment, the underlying architecture allows for future containerization and cloud-based scaling.

The PoC has removed the traditional limitations of Excel, such as worksheet size, formula depth, and fragile cross-sheet dependencies giving a basis for scalability over time, data reliability, and consistent performance, as the financial data grows. In addition to greater data volumes, the application supports more complex data models that are more structured, more object-oriented than traditional spreadsheets. This could include things like real estate records that could have fields for loan amount, terms of mortgage payments, and all of the payment history, or share transactions that could track dividends, market pricing, and country attribution. It is difficult to model the level of detail in Excel and to maintain its reliability. This limitation is also observed in studies of PFM tools, where traditional systems relying on manual entry and static formats often struggle to support growing data needs and complex financial contexts (Gafrikova et al. 2015).

One example of improved scalability is the automated net worth tracking feature. Previously, the stakeholder manually gathered asset values from various sources and combined them to update a dataset for visualization. In contrast, the PoC system calculates net worth automatically whenever any asset is modified and can also generate historical values retrospectively (see Figure 5).

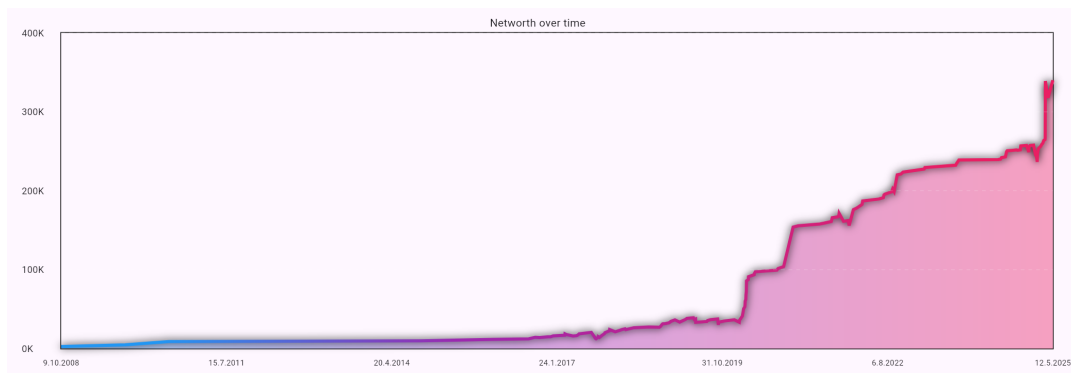


Figure 5: User's net worth over time

7.3 Automating financial data retrieval

One of the key goals identified during the problem definition phase was to reduce the manual effort required to update financial data, especially stock prices, exchange rates, and account balances. While full automation remains a challenge due to limited access to bank and broker APIs, the current solution introduces partial automation that meaningfully reduces the user's workload.

Stock prices are automatically retrieved through a Google Sheets integration, where the sheet is powered by the `GOOGLEFINANCE` function (See Figure 6). The application connects to this sheet through the Google Sheets API and fetches up-to-date prices for all tickers listed by the user. Although users must still maintain this ticker list and enter share transactions manually, the solution removes the need to repeatedly copy data between Excel files.

fx =GOOGLEFINANCE(B2)

A	B	C
Name	Ticker	Price
3M	MMM	134,32
Ares Capital	ARCC	19,9
Apple	AAPL	192,97
Coca-Cola Co	KO	71,15
iShares Core World	AMS:IWDA	87,69
iShares SP500	BIT:CSSPX	495,59
Elisa	ELISA	44,24
Google	GOOGL	156,36
Investor B	INVE-B	270,1
Johnson & Johnson	JNJ	149
Kesko	KESKOB	18,63
Microsoft	MSFT	385,55
NNN	NNN	38,94
Nordea	NDA-FI	10,57
Pepsi	PEP	143,23
Realty Income Corp	O	54,25
Sampo	SAMPO	8,44
Simon Property Group	SPG	146,81
Starbucks Corp	SBUX	83,85
Unilever	UNA	53,38
Visa	V	328,49
WP Carey	WPC	56,7

Figure 6: Google Sheet with the user's stock tickers powered by the GOOGLEFINANCE function

Exchange rates are retrieved daily using the HexaRate API and stored in the database for reuse. This ensures that all financial data regardless of original currency is converted consistently without requiring the user to look up or input rates manually.

While income and expense data are still entered manually through the application's user interface, the system centralizes this information for easier tracking. In the stakeholder's previous workflow, expense and income data were entered into Spendee, exported, and then incorporated into a financial report and master Excel file. In contrast, the new system decreases redundancy and improves consistency by maintaining all data in a centralized platform. This removes the need to update rates and prices across multiple tabs and formulas.

Future iterations may include integration with Open Banking APIs or broker-

specific services, if reliable and cost-effective options become available. The current solution offers a practical balance between automation, data accuracy, and user control. Achieving even partial automation has helped reduce friction for users and create a level of consistency. This aligns with findings by Cederberg (2013), who noted that users were more likely to adopt PFM tools when features such as transaction tracking and visual summaries were automated, reducing manual input and cognitive effort. However, as pointed by Kaye et al. (2014), some users may become frustrated with too much automation, such as inflexible or inaccurate expense categorization, which can hinder trust or usability. This supports the rationale for retaining user control alongside automation, ensuring the system remains both efficient and adaptable to individual preferences.

7.4 Centralized tracking across categories

The application provides structured and centralized tracking of financial data across asset and cash flow categories. Assets, including shares, cash, and real estate, are stored in dedicated collections and grouped using relevant identifiers to support organized analysis.

Shares are enriched with metadata such as category (e.g., dividend, growth), subcategory (e.g., REIT, BDC), and country of origin. These additional fields enable detailed breakdowns and are visualized through multi-layered pie charts, as shown in Figure 7.

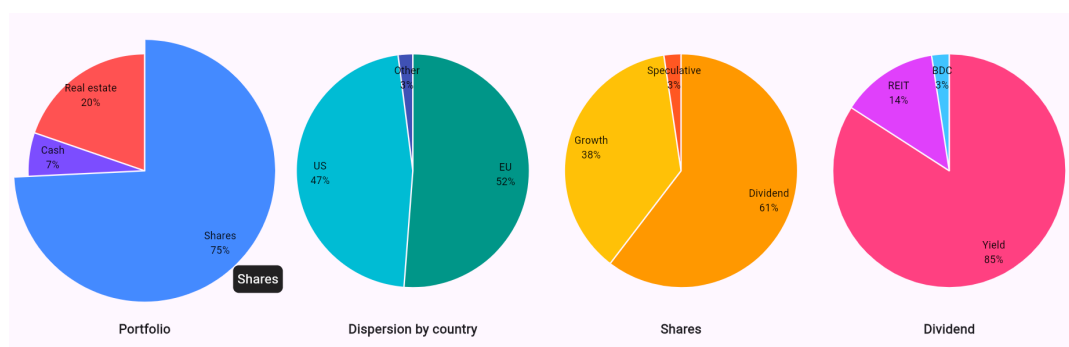


Figure 7: Share distribution visualized by category, country, and subcategory

Cash assets are grouped by account name (e.g., savings or investment account), while real estate holdings are categorized by property name. This helps users track the value and performance of each asset individually, without mixing different types of finances together (See Figure 8).

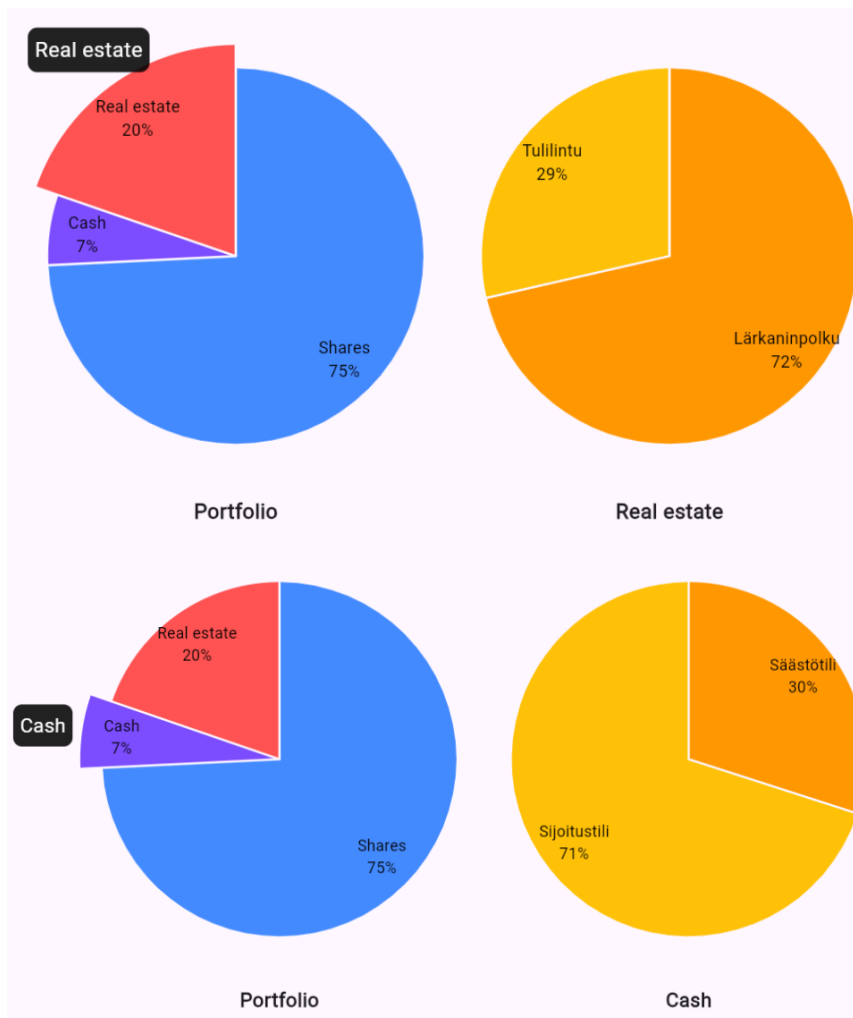


Figure 8: Other asset distribution visualized by category, country, and subcategory

Additionally, the application tracks income and expenses under the cash flow module. Transactions are tagged with labels such as food, salary, or shopping, enabling categorized insights. As shown in Figure 9, cash flow data is presented through categorized pie charts, a chronological transaction timeline, and a bar chart comparing monthly income and expenses.



Figure 9: Cash flow tracking: categorized income and expenses with monthly comparison

The system provides the user with insights that were previously fragmented throughout different tools and spreadsheets by standardizing categorization for both assets and transactions and consolidating all of the data within a single interface. This helps centralizing the user's financial management, and ultimately helps their decision-making process. More organized and categorized financial data can aid more informed insights and decision-making. This supports findings from Gafrikova et al. (2015), who described how PFM tools provide users with categorized expense tracking, grouped spending by type (e.g., rent, food, transportation), and aggregated financial information from various sources, which improves visibility into personal financial flows. Handayani et al. (2024) similarly noted that automatic expense trackers, categorization features, and integration across financial sources allow users to recognize their spending patterns and make better financial decisions.

7.5 Integrated and efficient reporting

One of the most labor-intensive tasks in the stakeholder's previous workflow was the creation of quarterly financial reports. This process involved manually copying charts, aggregating figures from various spreadsheets, and compiling them into a Word document. The PoC application replaces this with a fully integrated reporting feature that generates a comprehensive summary for a selected time period.

The report consolidates data from multiple parts of the system into a single PDF

file and includes the following elements:

- Monthly income, expenses, and surplus visualized in a grouped bar chart
- Savings percentage per month
- Expenses vs. income displayed as a line chart
- A detailed list of categorized cash flow transactions
- Net worth development over time and breakdown by asset category
- Share distribution by type, country, and subcategory

All data is automatically retrieved from the backend and visualized through charts and tables based on the selected time frame. The compiled report is exported as a downloadable PDF document, allowing users to archive or review their financial summaries without manual intervention.

This feature saves time and effort when generating reports on a recurring basis by replacing a multi-step manual process with one integrated functionality. As Cederberg (2013) says, automating repetitive financial tasks along with providing visual summaries for clarity can reduce the burden on users and ease adoption. Likewise, Shafiee et al. (2024) point out that reporting provides coherent and visual outputs to help establish understanding and inform decisions. It also satisfies the idea for an overview of finances by Kaye et al. (2014), consolidating a number of tools and processes into a singular reporting process.

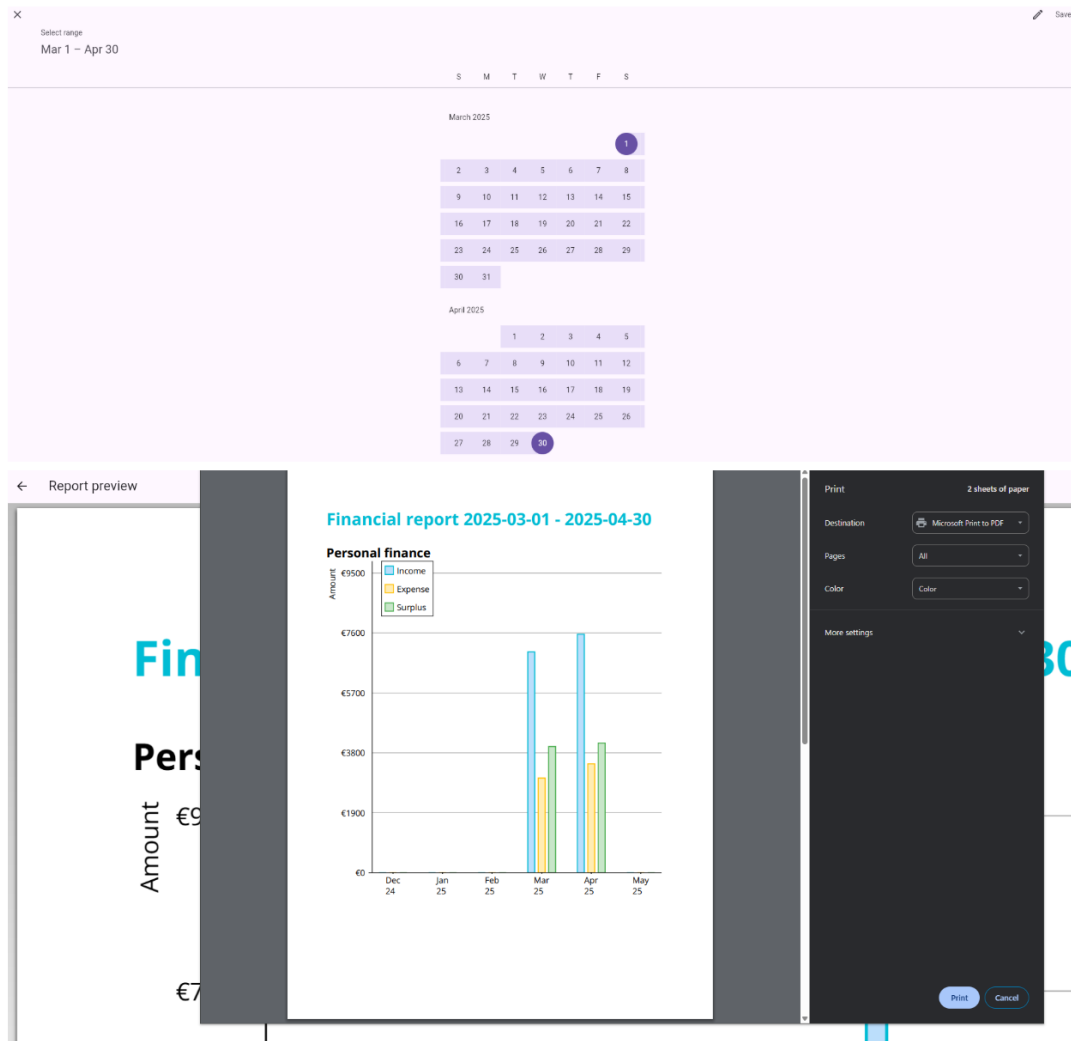


Figure 10: User-generated financial report for a selected time period

By automating the generation of structured and visually rich reports, the system supports more efficient and consistent financial review cycles while preserving the stakeholder's preference for detailed documentation and analysis.

8 Evaluation

This chapter presents the results of the evaluation phase. The goal of this phase is to assess how effectively the implemented artifact addresses the original problem and fulfills the defined solution objectives. Four evaluation methods were used: task-based user testing, a stakeholder feedback questionnaire, comparison to the baseline system, and analytical assessment against the design objectives.

8.1 Task-based user testing

Task-based user testing was conducted to observe the stakeholder interacting with the system in a realistic usage scenario. The purpose was to evaluate the usability, core functionality, and overall user experience. The stakeholder performed a predefined set of tasks on their own machine while being observed. Each task was assessed based on successful completion, workflow intuitiveness, and any usability issues or feedback that arose. The tasks were originally defined in Chapter 3.1.5.

All tasks were completed successfully. The stakeholder navigated the system confidently and required no external guidance to locate or use any of the key features. Task durations ranged from two to five minutes, depending on data entry requirements. For example:

- **Task 1:** The stakeholder imported share transactions using a CSV file and manually added two cash and two real estate transactions.
- **Task 2:** The user explored pie charts on the dashboard by category and reviewed data tables on the "Assets and Transactions" page, including testing the sorting and editing functionalities.
- **Task 3:** Multiple income and expense transactions were added across a four-month period, followed by a review of the updated cash flow visualizations.
- **Task 4:** The stakeholder generated a financial report for the period 1.1.2025–30.4.2025 and examined the contents of the generated PDF file.

Feedback during testing primarily concerned user interface clarity and usability improvements. Notable suggestions included:

- Enhancing tooltips in pie charts to show additional details (e.g., EUR value)
- Improving label clarity in the real estate table
- Adjusting the transaction ordering in the cash flow list for better intuitiveness

Despite these observations, all tasks were completed without errors or external assistance which indicates that the system provides a generally intuitive and user-

friendly experience. These findings are consistent with literature showing that structured, intuitive interfaces and visual feedback mechanisms play a big role in improving user engagement and confidence in personal finance tools. This is supported by Cederberg (2013), who found that simplified interfaces with visual summaries reduce user effort and promote adoption. Similarly, Shafiee et al. (2024) note that user-friendly and personalized interfaces improve users' understanding and confidence in financial decision-making.

8.2 Stakeholder insights from questionnaire

Following the task-based testing session, a structured questionnaire was administered to gather deeper feedback from the stakeholder. The questions were organized into four areas: general impressions, feature satisfaction, problem-solving impact, and suggestions for future development. This section summarizes the stakeholder's responses in each area.

8.2.1 General impressions

The stakeholder found the interface intuitive and user-friendly. Although some functionalities were not immediately visible, they were easily discovered through brief exploration. Overall, the structure and navigation of the application were considered effective and aligned with typical user expectations.

8.2.2 Feature satisfaction

The share tracking functionality was praised for its scalability and ease of use compared to the stakeholder's previous Excel system. The automation of data aggregation and value computation was seen as a major improvement.

The stakeholder found the cash flow feature useful and realistic but suggested clarifying its purpose whether it is meant for detailed budgeting like in apps such as Spendee, or for more general expense tracking. This would help determine whether the feature should be simplified or offer more customization.

Visualizations were described as clear and informative. The stakeholder highlighted the pie charts and trend graphs as particularly valuable for gaining quick insights. Minor refinements were suggested, particularly for the long-term capital development graph to improve clarity.

The PDF report generation feature was considered a strong improvement. It streamlined a previously manual and time-consuming reporting process, offering clear value through automation and integration.

8.2.3 Problem-solving impact

The stakeholder reported a reduction in manual work, especially with processes that went back and forth between excel and google sheets. The ability to see all the financial information in one interface was seen as a major improvement. This supports findings by Kaye et al. (2014) that users preferred simplified snapshots rather than a complete representation. In a similar manner, Cederberg (2013) noted that PFM systems offering centralized and easy-to-access financial data increase user motivation and ongoing engagement.

System stability and data reliability were also noteworthy. In contrast to the previous Excel workflow that was susceptible to broken formulas and inconsistencies, the PoC was praised as secure and dependable. Automatic integration of transactions and chronological organization contributed to a higher level of trust in the data. This finding supports the consideration of epistemic stability in personal finance tools, articulated by Herrala et al. (2023). They argued that users must be able to trust the logic and output of the application in order to experience the sense of empowerment and feeling of control over their finances. When tools are perceived as unpredictable or "black box", users' trust in the data and the system as a whole can decrease.

8.2.4 Suggestions and future improvements

Suggestions focused primarily on improving user experience and refining existing features. These included:

- Enhancing tooltip and graph clarity
- Adding the ability to modify transactions directly from the transaction list
- Clarifying the intended use of the cash flow functionality

Overall, the stakeholder expressed strong confidence in the application's potential. They indicated that with further refinement and iteration, the system could evolve into a fully deployable product that meets broader personal finance management needs.

8.3 Comparison to baseline

In order to assess the practical advantages of the implemented solution, a simple comparison was made between the stakeholder's existing financial tracking method to the PoC application. The original method consisted of numerous approaches and tools, such as Excel spreadsheets, Google Sheets, and the Spendee application, where consolidating all of the information was laborious and manual.

Table 9 summarizes key differences between the baseline setup and the PoC system.

Aspect	Old system (Baseline)	New system (PoC)
Tool fragmentation	Multiple tools: Excel, Spendee, Google Sheets	Centralized, web-based application
Share price updates	Manual copy-paste from Google Finance	Automated via Google Sheets API integration
Exchange rate updates	Manually checked and entered	Automatically fetched daily from HexaRate API
Income/expense tracking	Recorded in Spendee, analyzed in Excel	Entered and visualized within the application
Data visualization	Static, manually created Excel charts	Interactive, real-time visualizations and tables
Financial reporting	Manual report assembly in Word with Excel charts	Automated PDF reports with integrated data and visuals
Stability and accuracy	High risk of formula errors and broken references	Backend-calculated values with robust logic
Time spent updating	Significant manual effort and tool switching	Streamlined updates via integrated data flow

Table 9: Comparison of the baseline system and the PoC application

As seen in Table 9, the PoC solution meets the most important issues found from the previous system, by offering centralized data, an automated updates, and built-in analytic capabilities. In short, it minimizes user manual workload and error rates while increasing data transparency. The stakeholder was in agreement that the usability, automation, and trust in the accuracy of the data had improved. Torno et al. (2021) found that most personal finance solutions tend to focus on very few features or services, so this PoC focuses bridging these gaps.

8.4 Evaluation against design objectives

This section assesses how effectively the implemented PoC meets the functional and quality objectives defined in Chapter 5. Each requirement is reviewed based on the actual system implementation, stakeholder feedback, and outcomes observed during the evaluation phase.

FR-01: User interface (Desktop and mobile)

The system was implemented using Flutter, supporting deployment across desktop, web, and mobile platforms. The stakeholder confirmed the interface was intuitive and easy to use during desktop testing. However, the current implementation is primarily optimized for desktop and web, with mobile-specific layout adjustments still pending.
Status: Partially achieved

FR-02: Data visualization of capital and statistics

Interactive charts and graphs were implemented to visualize net worth trends, asset

categories, and cash flow summaries. The stakeholder highlighted these features as clear and significantly more effective than their previous manual visualizations. *Status: Achieved*

FR-03: Tagging

The data model supports asset classification by categories, subcategories, and country of origin. Tagging is applied consistently across shares, cash accounts, real estate, and cash flow transactions. *Status: Achieved*

QA-01: Correctness of statistics

Key calculations, including net worth, profit/loss, and savings rate, are performed by centralized backend logic, reducing risk of user error. The stakeholder confirmed the accuracy and reliability of the results. *Status: Achieved*

FR-04: Present real-time capital information

Real-time market data and exchange rates are fetched automatically, ensuring up-to-date capital values. This functionality was identified by the stakeholder as a major improvement over the manual update process. *Status: Achieved*

FR-05: Financial report generation

The system provides downloadable PDF reports summarizing key financial metrics over a selected time frame. This replaced the stakeholder's manual reporting process and was viewed as a major efficiency improvement. *Status: Achieved*

QA-02: Data access control

Although the data model is designed to support user-level data segregation, no authentication or authorization mechanisms were implemented. This limitation was acknowledged and deferred due to time constraints and PoC scope. *Status: Not achieved*

FR-06: Automatic data fetching

The system integrates with external sources to retrieve exchange rates and share prices. However, full automation of bank or broker transactions was not implemented. Despite this, the current solution meaningfully reduces manual effort. *Status: Partially achieved*

FR-07: Choose currency

All financial data is normalized to a base currency, with exchange rates fetched daily. This ensures consistency across calculations and supports user-defined currency views. *Status: Achieved*

FR-08: Customizable dashboards

Customizable dashboards were not implemented. All visualizations are static and predefined. This feature was deprioritized due to limited time and PoC focus. *Status: Not achieved*

FR-09: Manual financial data input

Structured forms and CSV import features were implemented for all major transaction types, enabling effective manual data entry and corrections. *Status: Achieved*

QA-03: Good response time

Performance during testing was consistently fast. Efficient data fetching, use of caching, and state management contributed to a responsive user experience. However no performance testing was conducted and some refinements would be needed for production version. *Status: Achieved*

In summary, all core "Must have" requirements were successfully implemented or partially fulfilled, ensuring that the fundamental goals of the PoC were met. Most "Should have" features were included, while lower-priority items such as authentication and dashboard customization were deferred for future development iterations.

9 Discussion

This chapter looks back on the project and reflects on what was achieved. It starts by reviewing how well the final solution addressed the original problem and what kind of feedback the stakeholder gave. It also considers how useful the Design Science Research Methodology (DSRM) was throughout the process. In addition, the chapter highlights some of the challenges faced during development and points out possible ways the solution could be improved or expanded in the future. Altogether, it gives a clear picture of what the project accomplished and where it could go next.

9.1 Summary of results and impact

The findings from this research project show that the PoC application accomplishes the identified goals defined earlier in the thesis. The core functionalities were implemented and tested with the actual end user in a realistic environment. User feedback confirmed that the application solved the primary stakeholder concerns: it reduced manual effort and mitigated human errors. This finding corresponds with Herrala et al. (2023)'s observation that complex in an error-prone tools can reduce financial confidence and raise unnecessary anxiety, especially for younger users. By reducing complexity and centralizing financial data, the PoC system improves both usability and reliability.

Unlike many existing PFM tools which focus on limited functionality, such as budgeting or tracking spending patterns (e.g., Mint, YNAB, Spendee), the PoC was designed to provide a more global and integrated approach across all aspects of PFM. The PoC provided a functionality for tracking investments, monitoring net worth, analyzing cash flow, and producing financial reports, in a more centralized way. The PoC's design directly addressed the fragmented focus of PFM as recognized by Torno et al. (2021). The study noted that most PFM apps focus on single service and rarely combine planning, tracking, and education in one app. While Kaye et al. (2014) noted that some tools provide discomfort due to perceived complexity, reliability, and fragmented view of PFM thus risking long-term engagement. The existing stakeholder system also lacked automation which was noted in the literature. According to Cederberg (2013), automating tasks and reducing manual user input were big motivators for adopting mobile financial tools, as they simplified workflows and reduced friction. The PoC application improves on this by providing automated retrieval of exchange rates, stock price updates, and aggregation of asset data in real-time.

One of the major strengths was the PoC being adaptable under the European context. Many of the popular PFM tools, such as Mint, Quicken, etc. are largely based on U.S. users, customarily linked to regionally-identified financial APIs. This also complements the findings of Stefanov et al. (2024), who stated that users from underrepresented markets, for example from Eastern Europe, face localization issues relating to global PFM tools. The stakeholder was also able to benefit from features that include multi-currency and share price updates for both Finnish and Swedish

stocks, which is commonly not available in mainstream financial applications.

Another identified core improvement was the transition on reporting from quarterly to real-time financial reporting. Previously, the stakeholder was only engaging in quarterly reviews due to time consuming process. The stakeholder can now analyze real-time dashboards in one quick view and initiate reports into pdf. This functionality supports the findings of Cederberg (2013), which implied that financial tools with less manual inputs and offered simplified workflows made it likely that users further engage with personal finance technologies. Similarly, Shafiee et al. (2024) noted that real-time, personalized outputs enhance users' understanding and support more responsive financial decision-making.

Finally, by delivering real-time insights and visual feedback through pie charts, bar graphs, and trend visualizations, the application promotes awareness and decision confidence. Bitrián et al. (2021) found that visual and gamified elements can increase users' perceived competence and engagement. Similarly, Shafiee et al. (2024) showed that personalized and real-time financial guidance helps users better understand complex financial factors and make more informed decisions.

9.2 Limitations of the solution

While the PoC successfully addressed core requirements, several limitations remain. Most unimplemented features were categorized as “nice to have” or optional and were intentionally excluded due to the limited scope and timeframe of the project. However, one critical weakness is the absence of user authentication and authorization. Security is a fundamental concern in financial technology, especially where sensitive personal data is involved. Kaye et al. (2014) explains that users are often reluctant to adopt digital finance tools due to fears around data access and privacy. Similarly, Handayani et al. (2024) identify security and data protection as central factors influencing user trust in fintech solutions. Herrala et al. (2023) further argue that when users feel a lack of control over their financial information, it can contribute to stress and disengagement, highlighting the importance of transparent and secure system design. These findings underscore the importance of implementing basic access control mechanisms in future iterations to strengthen system credibility and promote long-term use.

From a development perspective, the most significant limitation is the incomplete automation of financial data retrieval. Although the stakeholder expressed satisfaction with the current level of automation, the system lacks the scalability needed to support broader user base. Ideally, users would be able to connect their bank and brokerage accounts directly to the application, allowing automatic synchronization of transactions and balances. However, this level of integration remains difficult to implement due to different factors. Several stock price API's were investigated such as Finnhub, stockdata.org, eoddata, and Alpha vintage, but were eventually discarded due to limited support for European stocks or were too expensive. Investment brokers also do not have API's or they are limited, for example Nordnet offers API only in Sweden.

Additionally, from open-banking perspective, some options were investigated such as Yapily, Polygon, and GoCardless. These were not suitable for the context case as they were either targeted for enterprises or required a lot of development work. Another issue with the open-banking integrations is the automated categorization which was pointed out also by Kaye et al. (2014).

As a practical workaround, the PoC application integrates with Google Sheets using the `GOOGLEFINANCE` function to retrieve stock prices. This solution, while manual to set up, aligned with the stakeholder's workflow and allowed real-time pricing without additional financial costs. Still, the absence of seamless, API-based integration for broader asset classes and account types remains a major limitation. These findings support what Cederberg (2013) observed: manual data entry and scattered tools make it difficult for people to continue using personal finance systems in the long run.

Another thing to note is the stakeholders good financial literacy and therefore higher expectations from the system. As they have tracked their finances more than a decade, they have accumulated a lot of knowledge and experience in this domain. However, some usability improvements and educational content would be beneficial to add in the current application for more novice users. This need is also reflected in the literature. Torno et al. (2021) point out that few personal finance applications offer built-in financial education, despite its potential to increase trust and engagement. Similarly, Lasmiatun (2024) found that younger users often turn to digital platforms for financial guidance and prefer interactive and visual formats for learning. Kaye et al. (2014) also noted that many users do not engage with existing all-in-one financial systems, suggesting an opportunity to design more accessible and user-tailored financial experiences.

9.3 Reflection on DSRM

Design Science Research Methodology (DSRM) provided an effective structure for development of this project. The structured iterative process created a logical progression starting with a problem definition step and ending with an artifact evaluation step which was extremely useful in a solo development environment. DSRM consists of set of steps as described by Peffers et al. (2007) that enable researchers to develop solutions to real-world problems by the design and evaluation of artifacts.

Each phase of the DSRM directly informed the next. In Design and Development phase, the architectural decisions were derived from the functional and non-functional requirements. The requirements on the other hand were formatted based on the stakeholder needs found in the Problem identification phase. Evaluation phase benefited from the established objectives but also provided substance to their evaluation. This corresponds with the guidance of Hevner et al. (2004) that utility, quality, and efficacy are necessary criteria of evaluating design artifacts in information systems research context.

Having a real stakeholder involved made a big difference in the project. Their ongoing feedback gave practical insights that would have been hard to discover through theory alone. By taking part in the demonstration and evaluation, they ensured the solution met real needs and worked in practice. As De Sordi (2021) points out, involving practitioners in Design Science Research makes the results more relevant and strengthens the credibility of the research.

Overall, Design Science Research Methodology gave a clear but flexible way to work through the project. It made it easier to build a useful and practical solution while still following a solid research process. This made it a good choice for a project like this that combines real-world needs with academic work.

9.4 Opportunities for future development

The evaluation phase revealed several key opportunities to enhance the PoC application and move toward a production-ready solution. The most critical area for future development is the integration of automated data fetching from banks and brokers. Implementing such features would significantly reduce manual input, enhance user convenience, and improve the system's scalability for a wider audience. This addresses a limitation also observed by Torno et al. (2021) and Stefanov et al. (2024), who found that most PFM tools lack comprehensive integration across financial institutions, with Stefanov et al. (2024) further noting that many solutions offer only limited localized services, reducing their effectiveness in certain markets.

Usability improvements also emerged as a priority. The stakeholder identified a range of refinements that would improve the user experience, including the ability to edit transactions directly, more informative tooltips, and clearer differentiation between detailed budgeting and high-level cash flow tracking. This aligns with findings by Cederberg (2013), who described that users value intuitive, visually appealing interfaces that reduce manual effort and support financial self-awareness. Similarly, Bitrián et al. (2021) found that clear feedback mechanisms and well-designed user experiences foster perceived competence and autonomy which are two key drivers of sustained use in personal finance applications.

The addition of user authentication and authorization is essential for ensuring data privacy and security, especially if the system is to be deployed beyond the original stakeholder. On the technical side, the introduction of unit and integration testing, along with comprehensive documentation, would support long-term maintainability and facilitate collaboration in larger development teams. Also, splitting the architecture into several microservices would increase scalability and increase performance for a production environment where more traffic is to be expected.

10 Conclusions

This thesis covered the design and development of a centralized personal finance management application that addressed the problems faced by an individual stakeholder. The main issues involved the complexity, fragmentation, and manual workload that were found in the stakeholder's existing financial tracking setup. The system relied on a combination of Excel spreadsheets, third-party apps, and manual data consolidation. The objective was to create a customized solution that improves automation, visualization, and reporting while supporting personalized financial analysis.

The project followed the Design Science Research Methodology (DSRM), which provided a structured and iterative framework with six steps: problem identification, objective definition, design and development, demonstration, evaluation, and communication. This methodology supported both the technical execution and academic rigor of the research process. Each phase produced outcomes that could be used on the next stages, eventually resulting in a functioning proof-of-concept (PoC) application.

The final artifact includes asset tracking such as shares, cash, and real-estates, categorized cash flow monitoring, and financial PDF report generation. In addition to constant feedback during the development phase, these features were validated through task-based user testing and structured stakeholder feedback. Compared to the baseline system, the PoC significantly reduced manual effort, improved data reliability, and offered a centralized view of the user's financial status. These outcomes align with findings in the literature, which stressed that simplification, visual feedback, and trust are important for effective personal finance tools.

However, the PoC did not implement several planned features, including user authentication, customizable dashboards, or full data automation through banking and broker integrations. These limitations stemmed from external API constraints and the limited scope of a solo development project and would be main focus area if this application would be deployed to broader use.

One of the challenges faced during this thesis was finding good literature that really looked at how people manage their finances in everyday life. A lot of the existing research focuses on financial literacy, behavior, or planning theory, but there was not much detail about what tools people actually use. It was hard to find statistics or studies about which apps people rely on, what works for them, and what problems they run into. That is one of the main reasons this thesis came to life. Most apps on the market are built just for budgeting or tracking expenses. With simple Google search "personal finance apps," almost all of them fall into that category. But managing your finances is more than just budgeting. People also want to track investments, assets, and see their full financial picture in one place. More research is needed that looks into how people actually manage their finances and how tools could be built to support that.

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